Course and its outcomes

I Semester: Electromagnetism and Relativity (B010703T)

Unit I Guided electromagnetic waves: Transmission Lines and Wave Guides, Modes in a rectangular wave guide, Cavity resonators.

Unit II Tensor analysis: General coordinate transformation; contravariant, covariant and mixed tensors; metric tensor; raising and lowering of indices; contraction of indices.

Unit III Minkowsky space and Lorentz transformations: Geometry of space-time in Special Relativity; Minkowsky metric; Light cone and principle of causality; Invariance of Minkowsky metric under Lorentz transformations; Lorentz group; Proper, improper and orthochronous transformations; Pseudo-tensors.

Unit IV Covariant formulation of electromagnetism: Charge-current density four-vector; Scalar and Vector potentials; Gauge invariance; Electromagnetic potential four-vector; Electromagnetic field tensor; Lorentz transformation of electric and magnetic fields; Invariants of the electromagnetic field

Unit V Electromagnetic field of a charge moving with constant velocity, Covariant form of Lorentz force law; Dynamics of charged particles in static and uniform electric fields.

Reading List

1. Introduction to Electrodynamics, 4th edition, D. J. Griffiths (Pearson Education India Learning Private Limited; 2015)
2. A first Course in General Relativity, 2nd edition, Bernard Schutz (Cambridge University Press, 2009)
3. The Feynman Lectures on Physics, Vol. II: Mainly Electromagnetism and Matter, Richard Feynman, Robert B. Leighton, Matthew Sands (Pearson Education India, 2012)
4. Field and Wave Electromagnetics, 2nd edition, David K. Cheng (Pearson Education India, 2014)

Learning Outcomes

Knowledge:

The students

- will be familiar with the fundamental features and concepts of transmission lines and waveguides and their applications in real world including cavity resonators.
- will have basic understanding of tensor analysis.
- Will be able to explain the fundamental concepts of geometry of space time in special relativity and the principle of causality.
- will have knowledge about Lorentz group and electromagnetic field tensor.

Skills:

The students

- will be able to perform mathematical analysis of transmission lines and determine parametric conditions for better performance of transmission lines.
- Will be able to perform Lorentz transformation of electric and magnetic fields.
- Will be able to derive equation of motion of a charge particle and determine force on it when it being in static and uniform electric fields.